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**A Performance Benchmarking of Lightweight CNNs and Vision Transformers for
Automated Diabetic Retinopathy Grading Under Resource Constraints**

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Abstract

5 stages

Clinical DR severity scale

103,020

Augmented class-balanced images

98.91%

Top validation accuracy range
(MobileNetV3-Large)

98.1 %

EfficientNet-B0 achieved Acc.

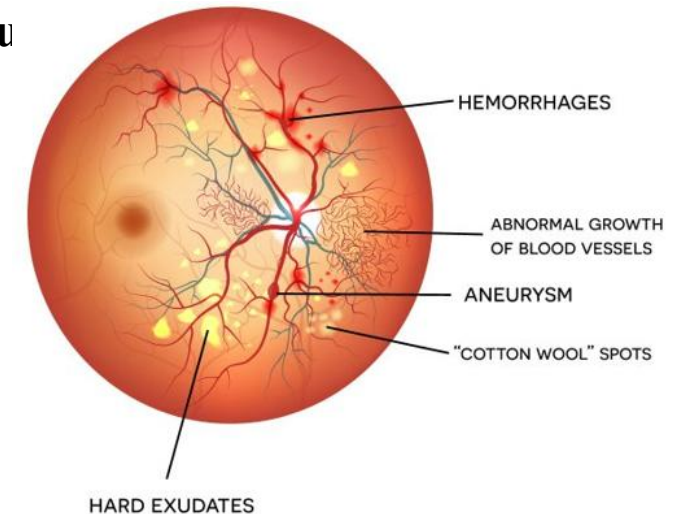
98.09 %

ViT achieved Acc.

- ❖ Diabetic Retinopathy is a major eye disease related to diabetes.
- ❖ In this research, we compare lightweight CNN models with Vision Transformer.
- ❖ We used 103,020 retinal images from EyePACS and APTOS datasets.
- ❖ MobileNetV3-Large achieved 98.91% accuracy for binary detection.
- ❖ ViT-Base achieved 98.09% accuracy .
- ❖ For five-class grading, EfficientNet-B0 achieved 98.1% accuracy.
- ❖ The main finding is that lightweight CNN models can provide high accuracy with fewer parameters.

Introduction

- ❖ **Diabetic Retinopathy (DR) is an eye problem from diabetes that damages the retina's blood vessels.**
- ❖ **It is one of the leading causes of blindness worldwide, especially in working-age adults.**
- ❖ **It has five severity stages, from No DR to Proliferative DR.**
- ❖ **Manual diagnosis is time-consuming and specialists are limited in many areas.**



Introduction (Cont.)

Problem :

- ❖ Large CNNs and ViT-Base (~86M parameters) require high computational resources.
- ❖ Their suitability for low-resource environments needs evaluation.

Objective :

- ❖ To Compare lightweight CNNs and ViT for DR detection and grading.
- ❖ To Evaluate accuracy and model size.
- ❖ To Identify models suitable for resource-limited clinical settings.

Literature Review

Previous CNN-Based Methods

- VGGNet, ResNet and DenseNet can achieve high accuracy.
- But they have a large number of parameters.
- MobileNetV2 and MobileNetV3 were developed for lightweight applications.
- EfficientNet improves the balance between accuracy and model size.

Vision Transformer (ViT)

- ViT uses self-attention.
- It can learn global relationships between image patches.
- Several studies reported good performance of ViT for retinal image analysis.
- However, ViT-Base has around 86M parameters.

Model Parameter Comparison

Approx. Parameters

MobileNetV3-Small	2.5M
MobileNetV2	3.5M
EfficientNet-B0	5.3M
MobileNetV3-Large	5.4M
DenseNet121	8.0M
ResNet50	25.6M
ViT-Base	86M
VGGNet16	138M

Literature Review (Cont.)

Research Gap :

GAP 01 Single DR Task Focus

Most studies focus on only one diabetic retinopathy (DR) problem at a time.

GAP 02 Binary vs Five-Class Grading

They rarely compare simple detection (Yes/No) with detailed 5-stage grading.

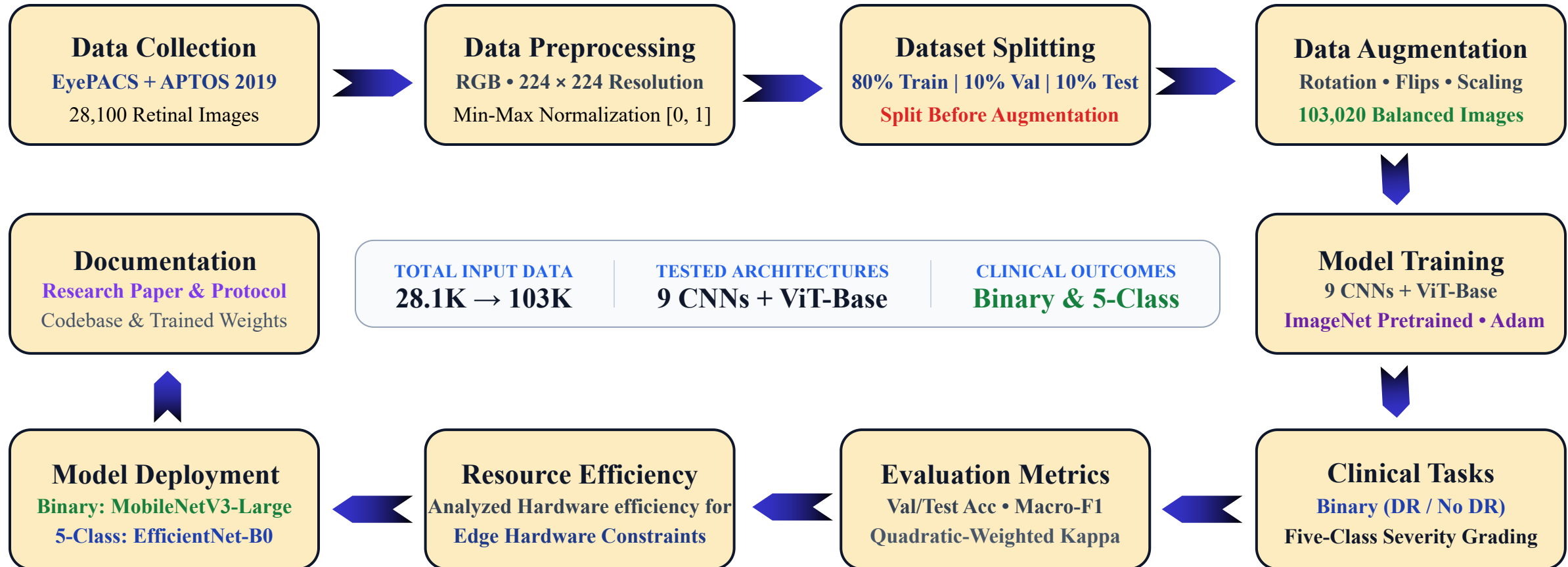
GAP 03 Accuracy vs Resource Efficiency

Most works only check accuracy, ignoring how fast or lightweight (resource-efficient) the model is.

GAP 04 ViT on Five-Class Grading

There is a lack of proper testing of Vision Transformers (ViT) on 5-stage DR classification.

Methodology



Proposed Work

Key Contribution :

A systematic resource-aware benchmark of lightweight CNNs and ViT for both binary DR detection and five-class DR severity grading.

DIMENSION 01



Binary Benchmark

MobileNetV3-Large vs. ViT-Base + other CNNs

DIMENSION 02



Five-Class Benchmark

8 CNN architectures for severity grading

DIMENSION 03



Efficiency Analysis

Accuracy compared with model parameter count

Innovation :



PILLAR 01

Binary + Five-Class DR evaluation in one benchmark



PILLAR 03

Accuracy-parameter efficiency analysis



PILLAR 02

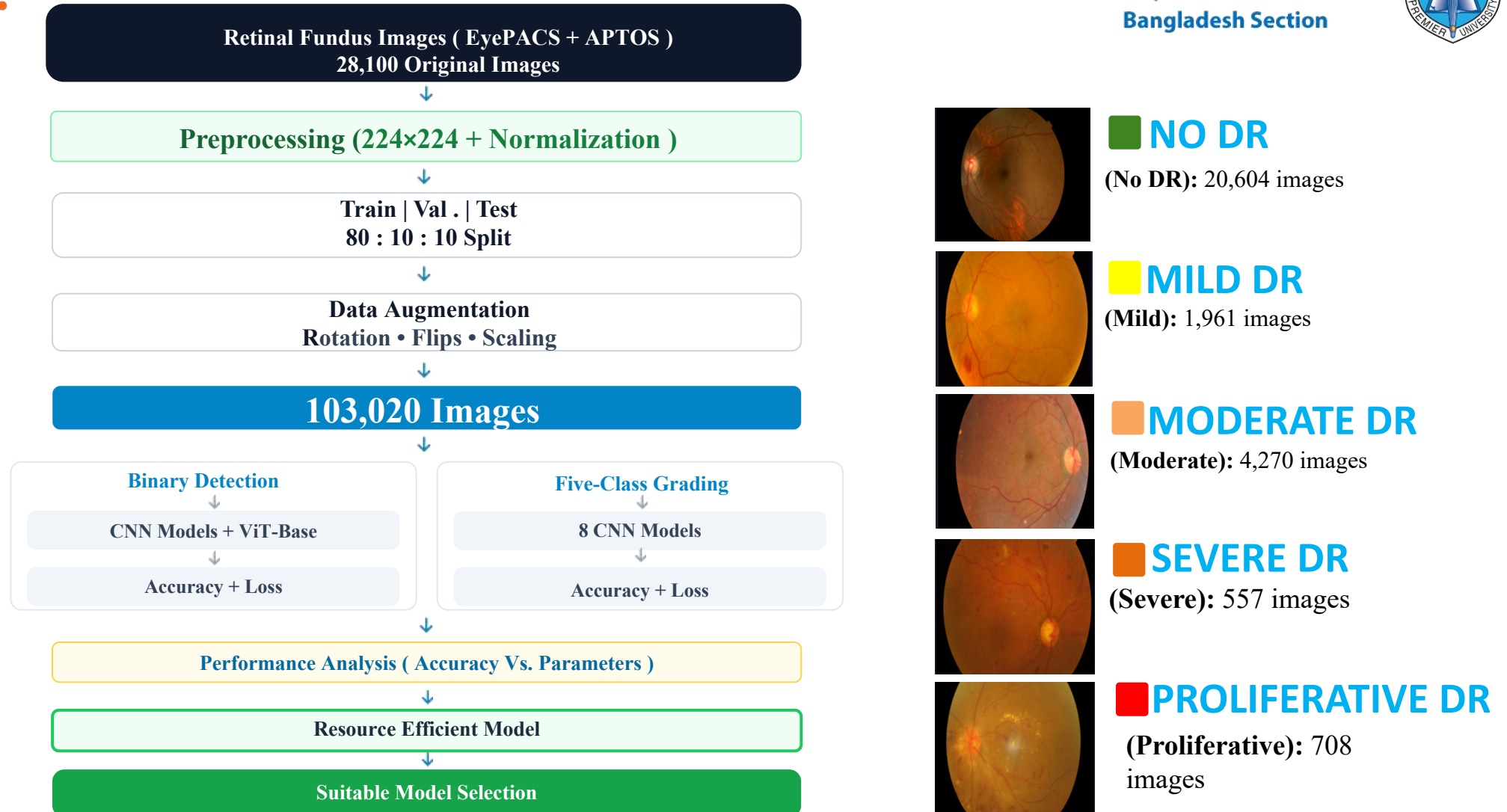
Lightweight CNNs compared with ViT-Base



PILLAR 04

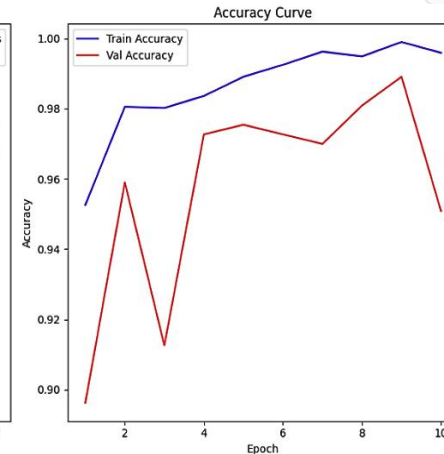
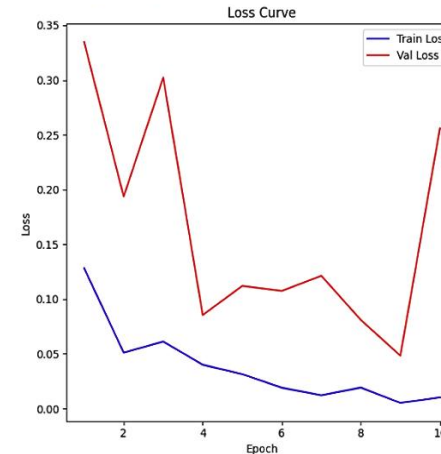
Resource-constrained deployment perspective

Proposed Work (Cont.)



Binary Diabetic Retinopathy Detection

Model	Type	Best Epoch	Train Acc.	Val. Loss ↓	Val. Acc. (%) ↑
★ MobileNetV3-Large	CNN	9/10	99.90%	0.0483	98.91%
★ MobileNetV2	CNN	3/10	98.50%	0.0713	98.91%
★ EfficientNet-B0	CNN	1/10	95.49%	0.0632	98.91%
DenseNet121	CNN	10/10	98.94%	0.0529	98.63%
MobileNetV3-Small	CNN	8/10	99.56%	0.0824	98.63%
ResNet50	CNN	5/10	98.67%	0.0640	98.36%
★ ViT-Base	Transformer	9/10	98.09%	0.0716	98.09%
VGGNet16	CNN	5/10	50.46%	0.6932	50.82%



MobileNetV3-Large

ViT

Binary DR Detection: CNN vs ViT

MobileNetV3-Large
98.91%
~5.4M Parameters

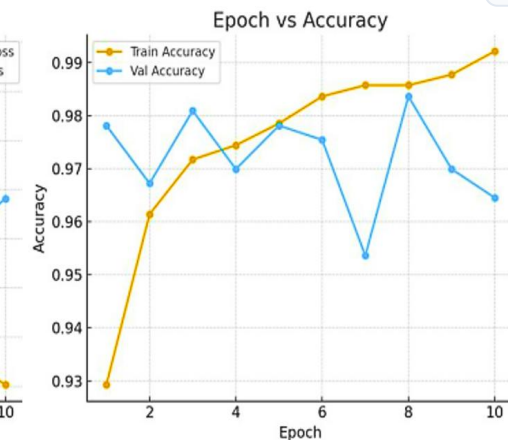
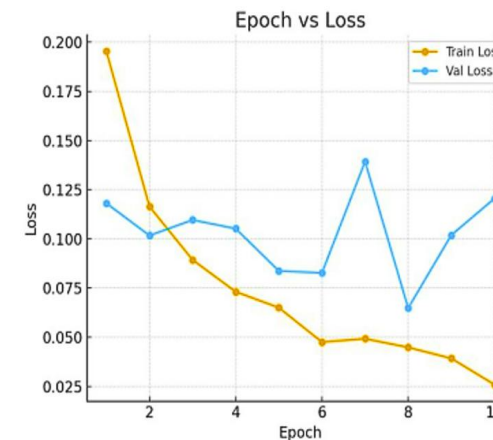
0.0483

MobileNetV3-Large
validation loss

ViT-Base
98.09%
86M Parameters

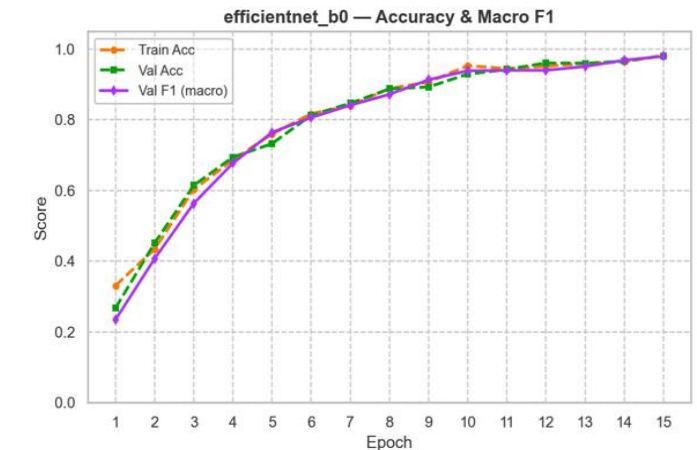
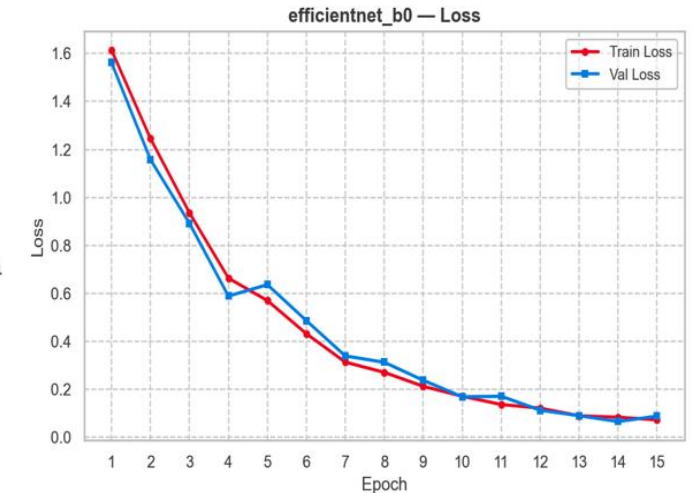
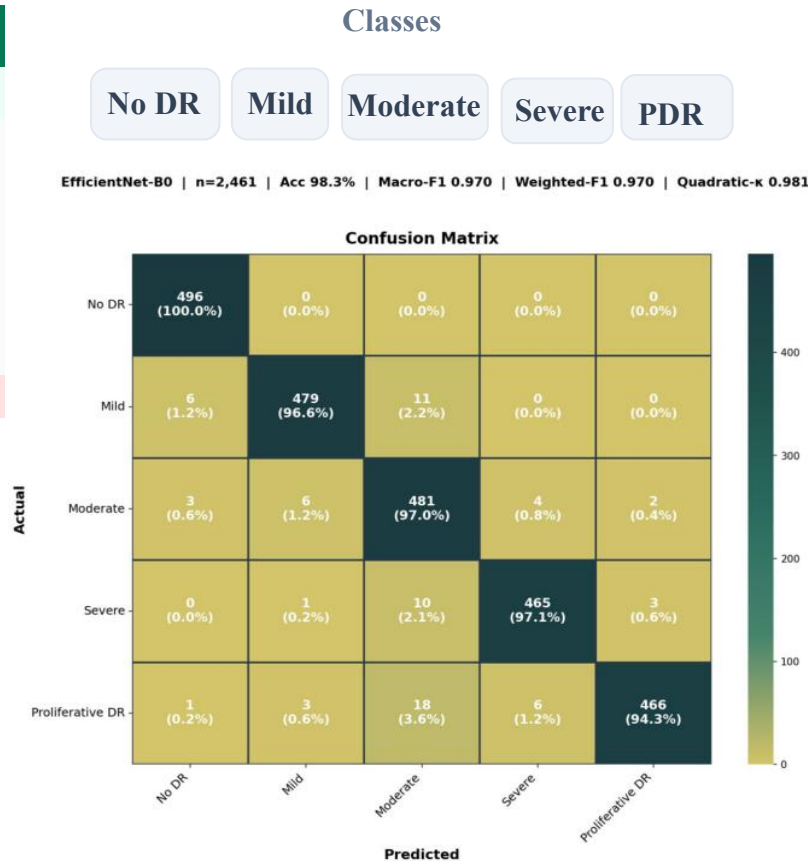
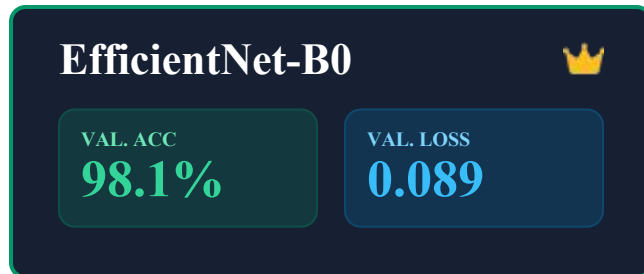
0.0716

ViT-Base validation loss



B. Five-Class Diabetic Retinopathy Grading

Model	Type	Epoch	Val. Loss ↓	Val. Acc. (%) ↑
EfficientNet-B0	Type	15	0.089	98.1%
DenseNet121	CNN	15	0.358	92.6%
MobileNetV3-Large	CNN	11	0.471	92.2%
Xception	CNN	12	0.449	91.3%
EfficientNet-B4	CNN	14	0.438	90.7%
ResNet50	CNN	12	0.493	90.4%
MobileNetV3-Small	CNN	14	0.558	88.3%
MobileNetV2	CNN	12	0.609	87.7%



Conclusion

- ☐ Lightweight CNNs performed very well for DR detection and grading.
- ☐ MobileNetV3-Large achieved 98.91% validation accuracy for binary detection.
- ☐ EfficientNet-B0 achieved 98.1% validation accuracy for five-class grading.
- ☐ EfficientNet-B0 achieved 98.3% test accuracy.
- ☐ Lightweight models can provide high accuracy with fewer parameters.
- ☐ The results support the use of lightweight CNNs for resource-constrained DR screening.

Future Work

- Test ViT for five-class DR grading.
- Compare more efficient Transformer models.
- Use k-fold cross-validation.
- Test on external datasets and different cameras .
- Add Grad-CAM or other explainability methods.
- Develop CNN-Transformer hybrid models.

Reference

- [1] International Diabetes Federation, IDF Diabetes Atlas, 10th ed. Brussels, Belgium: International Diabetes Federation, 2021.
- [2] D. A. Antonetti, R. Klein, and T. W. Gardner, "Diabetic retinopathy," *New England Journal of Medicine*, vol. 366, no. 13, pp. 1227–1239, Mar. 2012.
- [3] C. P. Wilkinson et al., "Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales," *Ophthalmology*, vol. 110, no. 9, pp. 1677–1682, Sep. 2003.
- [4] V. Gulshan et al., "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *JAMA*, vol. 316, no. 22, pp. 2402–2410, Dec. 2016.
- [5] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. ICLR*, 2015, pp. 1–14.
- [6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778.
- [7] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 4700–4708.
- [8] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017, pp. 1251–1258.
- [9] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 1314–1324.
- [10] A. G. Howard et al., "MobileNets: Efficient convolutional neural net works for mobile vision applications," *arXiv preprint arXiv:1704.04861*, 2017.

Thank You !